

CHEMISTRY STUDY GUIDE

Fundamental Concepts and Principles

1. Atomic Structure

Chemistry is the scientific study of matter, its properties, how and why substances combine or separate to form other substances, and how substances interact with energy.

Key Historical Developments:

- Dalton's Atomic Theory (1803): All matter composed of atoms
- Mendeleev's Periodic Table (1869): Organized elements by properties
- Bohr's Atomic Model (1913): Electrons in discrete energy levels

The atom consists of:

- Protons (+ charge): Located in nucleus, mass ≈ 1 amu
- Neutrons (neutral): Located in nucleus, mass ≈ 1 amu
- Electrons (- charge): Orbit nucleus, mass $\approx 1/1836$ amu

Atomic Number and Mass Number:

Mass Number (A) = Protons + Neutrons

Atomic Number (Z) = Number of Protons

Common Elements:

Element	Symbol	Atomic #	Atomic Mass
Hydrogen	H	1	1.008
Carbon	C	6	12.011
Oxygen	O	8	15.999
Sodium	Na	11	22.990
Chlorine	Cl	17	35.453

CHEMICAL BONDING & REACTIONS

2. Types of Chemical Bonds

Chemical bonds are forces that hold atoms together in molecules and compounds. Three primary types exist:

1. IONIC BONDS:

- Formed by complete transfer of electrons
- Between metals and non-metals
- Example: NaCl (Sodium Chloride)
- High melting/boiling points

2. COVALENT BONDS:

- Formed by sharing electron pairs
- Between non-metals
- Example: H₂O (Water), CH₄ (Methane)
- Can be polar or non-polar

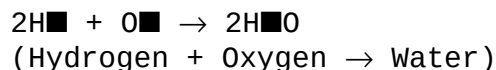
3. METALLIC BONDS:

- Sea of delocalized electrons
- Between metal atoms
- Example: Cu (Copper), Fe (Iron)
- Good conductors of heat/electricity
 - Atomic Radius: Increases ↓, decreases →
 - Ionization Energy: Increases →, decreases ↓
 - Electronegativity: Increases →, decreases ↓
 - Metallic Character: Increases ↓, decreases →
 - Electron Affinity: Generally increases →

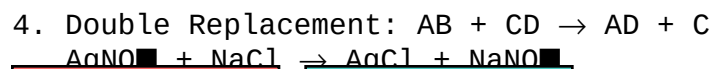
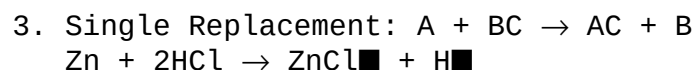
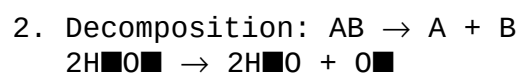
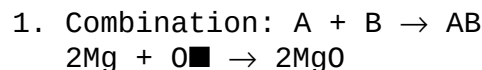
Periodic Table Trends:

3. Chemical Equations

Balanced Equation Example:



Types of Reactions:



Alkali
Metals

Halogens

4. Organic Chemistry Fundamentals

Organic chemistry studies carbon-containing compounds.

Carbon's unique properties:

- Forms 4 covalent bonds
- Can form chains, rings, and complex structures
- Tetravalent nature allows for isomers

Hydrocarbon Classes:

1. Alkanes: Single bonds (C-C), saturated
Example: Methane (CH₄), Ethane (C₂H₆)

2. Alkenes: Double bonds (C=C), unsaturated
Example: Ethene (C₂H₄)

3. Alkynes: Triple bonds (C≡C), unsaturated
Example: Ethyne (C₂H₂)

5. Biochemistry Essentials

4. Aromatic: Benzene Rings (C₆H₆)

Delocalized π electrons

MACROMOLECULES OF LIFE:

- Carbohydrates: Energy source (C₆H₁₂O₆ - Glucose)
- Lipids: Energy storage, cell membranes
- Proteins: Structure, enzymes (amino acid polymers)
- Nucleic Acids: Genetic information (DNA, RNA)

Key Biochemical Reactions:

Photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Cellular Respiration: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP}$

Hydrolysis: Breaking bonds with water

Dehydration Synthesis: Forming bonds, releasing water

Common Functional Groups:

Group	Formula	Example	Properties
Alcohol	-OH	C ₂ H ₅ OH	Polar, hydrogen bonding
Aldehyde	-CHO	CH ₃ CHO	Reactive, carbonyl
Ketone	C=O	CH ₃ COCH ₃	Solvent properties
Carboxylic Acid	-COOH	CH ₃ COOH	Acidic, hydrogen bonding
Amine	-NH ₂	CH ₃ NH ₂	Basic, forms salts

PHYSICAL CHEMISTRY

6. Solutions and Concentrations 7. Chemical Thermodynamics

Solution = Solute + Solvent

Concentration Units:

- Molarity (M) = moles solute / liters solution
- Molality (m) = moles solute / kg solvent
- Mass % = (mass solute / mass solution) × 100%
- ppm = (mass solute / mass solution) × 10⁶

Solubility Rules:

1. All Na⁺, K⁺, NH₄⁺ salts are soluble
2. All NO₃⁻, ClO₄⁻, ClO₃⁻ salts are soluble
3. Most Cl⁻, Br⁻, I⁻ salts soluble (except Ag⁺, Pb²⁺, Hg₂²⁺)
4. Most SO₄²⁻ salts soluble (except Ba²⁺, Pb²⁺, Ca²⁺)
5. Most OH⁻ salts insoluble (except Group 1, NH₄⁺, Ba²⁺)
6. Most S²⁻, CO₃²⁻, PO₄³⁻ salts insoluble

Laws of Thermodynamics:

1st Law: $\Delta U = Q - W$
(Energy conservation)

2nd Law: $\Delta S_{\text{universe}} \geq 0$
(Spontaneous processes increase disorder)

3rd Law: $S \rightarrow 0$ as $T \rightarrow 0$ K
(Perfect crystal at 0 K has zero entropy)

Gibbs Free Energy: $\Delta G = \Delta H - T\Delta S$

- $\Delta G < 0$: Spontaneous
- $\Delta G > 0$: Non-spontaneous
- $\Delta G = 0$: Equilibrium

8. Chemical Kinetics

Rate Laws:

For $aA + bB \rightarrow \text{products}$

$$\text{Rate} = k[A]^m[B]^n$$

Arrhenius Equation:

$$k = A \cdot e^{(-E_a/RT)}$$

where:

k = rate constant

A = frequency factor

E_a = activation energy

R = gas constant

T = temperature (K)

9. Gas Laws

Ideal Gas Law: $PV = nRT$

$$R = 0.0821 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K}$$

Combined Gas Law: $P_1V_1/T_1 = P_2V_2/T_2$

Dalton's Law: $P_{\text{total}} = \sum P_i$

Graham's Law: $\text{Rate}_1 / \text{Rate}_2 = \sqrt{(M_2 / M_1)}$

Van der Waals Equation:

$$[P + a(n/V)^2](V - nb) = nRT$$

Key Takeaways:

1. Chemistry connects microscopic particles to macroscopic properties
2. Periodic trends help predict element behavior
3. Bond type determines compound properties
4. Organic compounds form basis of life
5. Energy changes drive chemical reactions
6. Equilibrium balances forward and reverse reactions

Chemistry is the science of matter and the changes it undergoes